Nexus LMS — Complete Schemas & Flowcharts

Unified Technical Architecture Blueprint, Relational Schemas (24 Tables), and System Flow Diagrams

Document Version: 3.2.0 Enterprise	**Database Engine:** SQLite / PostgreSQL Dual-Compatible	**Total Tables:** 24 Relational Entities
Live Cloudflare URL: boats-revisions-released-strap.trycloudflare.com	**Local Server:** http://127.0.0.1:8000	**Repository:** github.com/rishrivastava/nexus-lms

1. System Architectural Topology & ER Relationships

Nexus LMS is structured into 6 decoupled architectural domains centered around high throughput, data integrity, and strict departmental governance. The diagram below illustrates how core entities connect across administrative boundaries:

Domain 1: Identity & Governance users • departments • teacher_departments • dashboard_permissions • audit_logs	→ ***1:N Mapping*** →	**Domain 2: Academic Curriculum** courses • modules • lessons • enrollments • lesson_progress
Domain 3: Subjective Answer Sheets answer_sheet_submissions (Handwritten PDF copies • Faculty Marks • Admin Surveillance)	↔ ***Course Context*** ↔	**Domain 4: Live Classes & Doubts** live_classes • live_class_questions • support_queries (Real-time Q&A; Queue)
Domain 5: Cognitive Assessments & AI quizzes • quiz_questions • quiz_attempts • ai_usage_logs (Gemini LLM)	→ ***Completion*** →	**Domain 6: Certification & Mentoring** certificates (Tamper-proof QR • Unique IDs) • mentoring_sessions (1-on-1)

2. Core Operational Workflow Diagrams

Below are the primary operational lifecycles implemented in Nexus LMS, documenting data flow, state transitions, security checkpoints, and user roles.

Workflow 2.1: Multi-Department Faculty Governance & Course Assignment Flow

Step 1: Multi-Dept Assignment	Step 2: Department Filter	Step 3: Course Assignment	Step 4: Class Isolation
Admin maps teacher to 1 or more departments via POST /department/teachers/assign-departments. *Records stored in teacher_departments.*	System fetches assignable courses via GET /api/teachers/{id}/assignable-courses. **Strict Filter:** Only courses belonging to teacher's departments returned.	Admin assigns course via POST /department/teachers/assign-course. **Server Validation:** Rejects if teacher not in course department with 400 error.	Teacher logs in: Teacher Studio (/teacher), Dashboard, and Profile strictly show **only explicitly assigned classes**.

Workflow 2.2: Subjective Ans Sheet Upload & Administrative Surveillance Window

Stage A: Student Upload	Stage B: Teacher Profile Alert	Stage C: Faculty Evaluation	Stage D: Admin Surveillance
Student submits handwritten PDF answer copy for course (POST /api/course/{id}/upload-answer-sheet). Status: submitted.	Answer PDF automatically populates on the course teacher's **Profile section** under *'Enrolled Student Answer Sheets & Evaluations'*.	Teacher reviews PDF, awards marks, and writes qualitative feedback (POST /api/answer-sheet/{id}/evaluate). Status: evaluated.	Admin monitors live metrics on /department?tab=answer-sheets: **Total Received, Evaluated, Pending, and Completion Rate %**.

Workflow 2.3: Student 360° Inspection & Mylo AI (Google Gemini) Generation

Student 360° Portfolio Modal	Mylo AI Quiz Synthesis	Live Classes & Doubt Queue	1-on-1 Mentorship Booking
Faculty clicks *'View Student Profile'* in Teacher Studio → /api/students/{id}/profile-modal displays institutional data, UDISE code, course completion %, and answer copies.	Teacher triggers AI quiz generation → Gemini synthesizes Bloom's taxonomy MCQs → JSON validated → attached as lesson quiz module.	Faculty creates live session → students stream video → in-stream doubt queue allows real-time questions → teacher answers on stream.	Student books counseling appointment → faculty sets Meet link → 1-on-1 session held → faculty logs outcome notes.

3. Complete Database Schema Reference (All 24 Relational Entities)

Below is the exact SQL relational schema for every database entity in Nexus LMS, documenting field names, data types, primary keys, nullability, defaults, and foreign key constraints.

Table ai_usage_logs — Token metrics, prompt type, latency, and estimated cost tracking for Google Gemini AI

#	Column Name	Type	Key / Null	Default
0	**id**	INTEGER	**PK**	—
1	**user_id**	INTEGER	NULL	—
2	**user_name**	TEXT	NULL	—
3	**user_role**	TEXT	NULL	—
4	**course_id**	INTEGER	NULL	—
5	**course_title**	TEXT	NULL	—
6	**subject**	TEXT	NULL	—
7	**feature_type**	TEXT	NOT NULL	—
8	**prompt_tokens**	INTEGER	NULL	0
9	**completion_tokens**	INTEGER	NULL	0
10	**total_tokens**	INTEGER	NULL	0
11	**estimated_cost_usd**	REAL	NULL	0.0
12	**created_at**	TIMESTAMP	NULL	CURRENT_TIMESTAMP

Table answer_sheet_submissions — Subjective handwritten PDF uploads, marks awarded, evaluator ID, feedback remarks

#	Column Name	Type	Key / Null	Default
0	**id**	INTEGER	**PK**	—
1	**user_id**	INTEGER	NOT NULL	—
2	**course_id**	INTEGER	NOT NULL	—
3	**module_id**	INTEGER	NULL	—
4	**lesson_id**	INTEGER	NULL	—
5	**file_path**	TEXT	NOT NULL	—
6	**original_filename**	TEXT	NOT NULL	—
7	**remarks**	TEXT	NULL	—
8	**status**	TEXT	NULL	'submitted'
9	**marks_awarded**	REAL	NULL	—
10	**max_marks**	REAL	NULL	20.0
11	**feedback**	TEXT	NULL	—
12	**evaluated_by**	INTEGER	NULL	—
13	**evaluated_at**	TIMESTAMP	NULL	—
14	**created_at**	TIMESTAMP	NULL	CURRENT_TIMESTAMP

Table audit_logs — Security audit trail recording user logins, course modifications, and faculty reassignments

#	Column Name	Type	Key / Null	Default
0	**id**	INTEGER	**PK**	—
1	**user_id**	INTEGER	NULL	—
2	**action**	TEXT	NOT NULL	—
3	**entity_type**	TEXT	NOT NULL	—
4	**entity_id**	TEXT	NULL	—
5	**details**	TEXT	NULL	—
6	**created_at**	TIMESTAMP	NULL	CURRENT_TIMESTAMP

Table certificates — Tamper-proof completion credentials, unique identifier, QR code data, and revocation audit

#	Column Name	Type	Key / Null	Default
0	**id**	INTEGER	**PK**	—
1	**certificate_id**	TEXT	NOT NULL	—
2	**student_id**	INTEGER	NOT NULL	—
3	**student_name**	TEXT	NOT NULL	—
4	**student_uid**	TEXT	NOT NULL	—
5	**course_id**	INTEGER	NOT NULL	—
6	**course_name**	TEXT	NOT NULL	—
7	**course_code**	TEXT	NULL	—
8	**issued_by**	TEXT	NOT NULL	'Leadership Boulevard Pvt Ltd'
9	**issued_at**	TIMESTAMP	NULL	CURRENT_TIMESTAMP
10	**status**	TEXT	NOT NULL	'verified'
11	**revoked_at**	TIMESTAMP	NULL	—
12	**revocation_reason**	TEXT	NULL	—
13	**reissued_from_id**	TEXT	NULL	—
14	**reissued_to_id**	TEXT	NULL	—
15	**qr_code_data_uri**	TEXT	NULL	—
16	**verification_url**	TEXT	NULL	—
17	**verification_count**	INTEGER	NULL	0
18	**last_verified_at**	TIMESTAMP	NULL	—
19	**completion_date**	TEXT	NULL	—
20	**grade_score**	REAL	NULL	100.0

Table courses — Course master catalog, code, assigned teacher_id, department, subject, publication status

#	Column Name	Type	Key / Null	Default
0	**id**	INTEGER	**PK**	—
1	**title**	TEXT	NOT NULL	—
2	**code**	TEXT	NOT NULL	—
3	**department**	TEXT	NOT NULL	—
4	**description**	TEXT	NULL	—
5	**thumbnail_url**	TEXT	NULL	—
6	**teacher_id**	INTEGER	NOT NULL	—
7	**is_published**	INTEGER	NULL	1
8	**created_at**	TIMESTAMP	NULL	CURRENT_TIMESTAMP
9	**subject**	TEXT	NULL	—
10	**status**	TEXT	NULL	'published'
11	**archived_at**	TIMESTAMP	NULL	—
12	**deleted_at**	TIMESTAMP	NULL	—
13	**subtitle**	TEXT	NULL	—
14	**learning_path**	TEXT	NULL	—
15	**instructions**	TEXT	NULL	—

Table dashboard_permissions — Role-based widget access controls for administrative dashboards

#	Column Name	Type	Key / Null	Default
0	**widget_key**	TEXT	**PK**	—
1	**widget_title**	TEXT	NOT NULL	—
2	**allowed_roles**	TEXT	NOT NULL	—
3	**is_enabled**	INTEGER	NULL	1

Table departments — Academic buckets, branding, welcome announcement, and ans_writing_enabled toggle

#	Column Name	Type	Key / Null	Default
0	**id**	INTEGER	**PK**	—
1	**name**	TEXT	NOT NULL	—
2	**code**	TEXT	NOT NULL	—
3	**description**	TEXT	NULL	—
4	**banner_url**	TEXT	NULL	—
5	**branding_tagline**	TEXT	NULL	'transforming education'
6	**welcome_message**	TEXT	NULL	—
7	**created_at**	TIMESTAMP	NULL	CURRENT_TIMESTAMP
8	**ans_writing_enabled**	INTEGER	NULL	1

Table enrollments — Student course enrollment registry and completion status

#	Column Name	Type	Key / Null	Default
0	**id**	INTEGER	**PK**	—
1	**user_id**	INTEGER	NOT NULL	—
2	**course_id**	INTEGER	NOT NULL	—
3	**enrolled_at**	TIMESTAMP	NULL	CURRENT_TIMESTAMP
4	**completed_at**	TIMESTAMP	NULL	—

Table lesson_progress — Granular lesson completion status and timestamp for student velocity

#	Column Name	Type	Key / Null	Default
0	**id**	INTEGER	**PK**	—
1	**user_id**	INTEGER	NOT NULL	—
2	**lesson_id**	INTEGER	NOT NULL	—
3	**is_completed**	INTEGER	NULL	0
4	**watched_seconds**	INTEGER	NULL	0
5	**completed_at**	TIMESTAMP	NULL	—

Table lessons — Individual lesson elements (YouTube video, lecture PDF document, or interactive Quiz)

#	Column Name	Type	Key / Null	Default
0	**id**	INTEGER	**PK**	—
1	**module_id**	INTEGER	NOT NULL	—
2	**title**	TEXT	NOT NULL	—
3	**content_type**	TEXT	NOT NULL	—
4	**content_value**	TEXT	NULL	—
5	**position**	INTEGER	NULL	1
6	**duration_minutes**	INTEGER	NULL	10
7	**subtitle**	TEXT	NULL	—
8	**description**	TEXT	NULL	—

Table live_class_questions — In-stream real-time student doubt submissions with teacher moderation resolution

#	Column Name	Type	Key / Null	Default
0	**id**	INTEGER	**PK**	—
1	**live_class_id**	INTEGER	NOT NULL	—
2	**user_id**	INTEGER	NOT NULL	—
3	**user_name**	TEXT	NOT NULL	—
4	**question_text**	TEXT	NOT NULL	—
5	**is_answered**	INTEGER	NULL	0
6	**answer_text**	TEXT	NULL	—
7	**created_at**	TIMESTAMP	NULL	CURRENT_TIMESTAMP

Table live_classes — Scheduled, live, and completed streaming sessions with Google Meet/WebRTC URLs

#	Column Name	Type	Key / Null	Default
0	**id**	INTEGER	**PK**	—
1	**course_id**	INTEGER	NOT NULL	—
2	**teacher_id**	INTEGER	NOT NULL	—
3	**title**	TEXT	NOT NULL	—
4	**description**	TEXT	NULL	—
5	**platform**	TEXT	NOT NULL	'zoom'
6	**meeting_url**	TEXT	NOT NULL	—
7	**meeting_id**	TEXT	NULL	—
8	**meeting_passcode**	TEXT	NULL	—
9	**scheduled_at**	TIMESTAMP	NOT NULL	—
10	**duration_minutes**	INTEGER	NULL	60
11	**status**	TEXT	NULL	'scheduled'
12	**recording_url**	TEXT	NULL	—
13	**created_at**	TIMESTAMP	NULL	CURRENT_TIMESTAMP

Table mentoring_sessions — 1-on-1 faculty counseling appointments, scheduled date/time, Meet link, status, and notes

#	Column Name	Type	Key / Null	Default
0	**id**	INTEGER	**PK**	—
1	**initiator_role**	TEXT	NOT NULL	—
2	**student_id**	INTEGER	NOT NULL	—
3	**teacher_id**	INTEGER	NULL	—
4	**course_id**	INTEGER	NULL	—
5	**topic**	TEXT	NOT NULL	—
6	**description**	TEXT	NULL	—
7	**preferred_date**	TEXT	NULL	—
8	**preferred_time**	TEXT	NULL	—
9	**scheduled_date**	TEXT	NULL	—
10	**scheduled_time**	TEXT	NULL	—
11	**meeting_link**	TEXT	NULL	—
12	**status**	TEXT	NULL	'pending_admin_assignment'
13	**student_notes**	TEXT	NULL	—
14	**teacher_notes**	TEXT	NULL	—
15	**created_at**	TIMESTAMP	NULL	CURRENT_TIMESTAMP
16	**updated_at**	TIMESTAMP	NULL	CURRENT_TIMESTAMP

Table modules — Curriculum units belonging to a course (ordered by position)

#	Column Name	Type	Key / Null	Default
0	**id**	INTEGER	**PK**	—
1	**course_id**	INTEGER	NOT NULL	—
2	**title**	TEXT	NOT NULL	—
3	**position**	INTEGER	NULL	1
4	**description**	TEXT	NULL	—

Table notification_settings — User communication preferences for email and in-app alert categories

#	Column Name	Type	Key / Null	Default
0	**event_type**	TEXT	**PK**	—
1	**event_title**	TEXT	NOT NULL	—
2	**is_enabled**	INTEGER	NULL	1
3	**target_roles**	TEXT	NOT NULL	—

Table notifications — In-app alerts for course assignments, evaluation completion, and live class reminders

#	Column Name	Type	Key / Null	Default
0	**id**	INTEGER	**PK**	—
1	**user_id**	INTEGER	NOT NULL	—
2	**title**	TEXT	NOT NULL	—
3	**message**	TEXT	NOT NULL	—
4	**link_url**	TEXT	NULL	—
5	**category**	TEXT	NULL	'general'
6	**is_read**	INTEGER	NULL	0
7	**created_at**	TIMESTAMP	NULL	CURRENT_TIMESTAMP

Table quiz_attempts — Student quiz submission records, percentage score, passed/failed boolean, time spent

#	Column Name	Type	Key / Null	Default
0	**id**	INTEGER	**PK**	—
1	**user_id**	INTEGER	NOT NULL	—
2	**quiz_id**	INTEGER	NOT NULL	—
3	**score_percent**	REAL	NOT NULL	—
4	**passed**	INTEGER	NOT NULL	—
5	**attempted_at**	TIMESTAMP	NULL	CURRENT_TIMESTAMP
6	**started_at**	TIMESTAMP	NULL	—
7	**time_taken_seconds**	INTEGER	NULL	0
8	**attempt_data**	TEXT	NULL	—

Table quiz_questions — Assessment multiple-choice items, options A-D, correct answer key, and pedagogical rationale

#	Column Name	Type	Key / Null	Default
0	**id**	INTEGER	**PK**	—
1	**quiz_id**	INTEGER	NOT NULL	—
2	**question_text**	TEXT	NOT NULL	—
3	**option_a**	TEXT	NOT NULL	—
4	**option_b**	TEXT	NOT NULL	—
5	**option_c**	TEXT	NOT NULL	—
6	**option_d**	TEXT	NOT NULL	—
7	**correct_option**	TEXT	NOT NULL	—
8	**explanation**	TEXT	NULL	—

Table quizzes — Assessment configuration, passing threshold percentage, time limit in minutes

#	Column Name	Type	Key / Null	Default
0	**id**	INTEGER	**PK**	—
1	**lesson_id**	INTEGER	NOT NULL	—
2	**passing_score**	INTEGER	NULL	70
3	**created_at**	TIMESTAMP	NULL	CURRENT_TIMESTAMP
4	**time_limit_minutes**	INTEGER	NULL	10

Table student_conflicts — Conflict resolution queue for handling duplicate national student registrations

#	Column Name	Type	Key / Null	Default
0	**id**	INTEGER	**PK**	—
1	**student_name**	TEXT	NOT NULL	—
2	**guardian_name**	TEXT	NOT NULL	—
3	**existing_user_id**	TEXT	NOT NULL	—
4	**requested_data**	TEXT	NULL	—
5	**status**	TEXT	NULL	'pending'
6	**created_at**	TIMESTAMP	NULL	CURRENT_TIMESTAMP

Table support_queries — Contextual support tickets, instant Gemini AI suggestions, and faculty resolution

#	Column Name	Type	Key / Null	Default
0	**id**	INTEGER	**PK**	—
1	**ticket_id**	TEXT	NOT NULL	—
2	**user_id**	TEXT	NOT NULL	—
3	**user_role**	TEXT	NOT NULL	—
4	**user_name**	TEXT	NOT NULL	—
5	**user_email**	TEXT	NULL	—
6	**course_id**	INTEGER	NULL	—
7	**course_title**	TEXT	NULL	—
8	**subject**	TEXT	NOT NULL	—
9	**category**	TEXT	NULL	'General'
10	**message**	TEXT	NOT NULL	—
11	**ai_response**	TEXT	NULL	—
12	**admin_reply**	TEXT	NULL	—
13	**replied_by**	TEXT	NULL	—
14	**replied_at**	TIMESTAMP	NULL	—
15	**status**	TEXT	NULL	'open'
16	**visibility**	TEXT	NOT NULL	—
17	**created_at**	TIMESTAMP	NULL	CURRENT_TIMESTAMP

Table system_settings — Global system configurations (branding, Mylo AI toggles, institution name)

#	Column Name	Type	Key / Null	Default
0	**key**	TEXT	**PK**	—
1	**value**	TEXT	NOT NULL	—

Table teacher_departments — Relational link enabling multi-department governance and course assignment verification

#	Column Name	Type	Key / Null	Default
0	**id**	INTEGER	**PK**	—
1	**teacher_id**	INTEGER	NOT NULL	—
2	**department_id**	INTEGER	NOT NULL	—
3	**created_at**	TIMESTAMP	NULL	CURRENT_TIMESTAMP

Table users — Primary accounts, auth hash, role (admin/teacher/student), national K-12 data (UDISE, district, guardian)

#	Column Name	Type	Key / Null	Default
0	**id**	INTEGER	**PK**	—
1	**user_id**	TEXT	NOT NULL	—
2	**full_name**	TEXT	NOT NULL	—
3	**department**	TEXT	NOT NULL	—
4	**email**	TEXT	NULL	—
5	**password_hash**	TEXT	NOT NULL	—
6	**role**	TEXT	NOT NULL	'student'
7	**must_change_password**	INTEGER	NULL	1
8	**created_at**	TIMESTAMP	NULL	CURRENT_TIMESTAMP
9	**guardian_name**	TEXT	NULL	—
10	**school_name**	TEXT	NULL	—
11	**udise_code**	TEXT	NULL	—
12	**block**	TEXT	NULL	—
13	**district**	TEXT	NULL	—
14	**state**	TEXT	NULL	—
15	**pincode**	TEXT	NULL	—
16	**last_active_at**	TIMESTAMP	NULL	—
17	**normalized_name**	TEXT	NULL	—
18	**normalized_guardian**	TEXT	NULL	—
19	**designation**	TEXT	NULL	—
20	**avatar_url**	TEXT	NULL	—
21	**phone**	TEXT	NULL	—
22	**bio**	TEXT	NULL	—